ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CCS2132 PROGRAMMING ESSENTIALS IN PYTHON

DEFERED EXAMINATION

DATE: 19TH DECEMBER, 2024

TIME: 09:00 – 12:00 HOURS

TIME ALLOWED: WRITING – THREE HOURS: READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: this question is **compulsory** and must be attempted.
2. Sections B: Answer **THREE (3)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

a) Imagine you are working as a financial consultant for a microfinance institution. Your team has been tasked with developing a tool to help clients understand their loan repayment obligations before taking a loan. One of the key requirements is to calculate the total repayment amount based on the following inputs:

- Calculate both **simple interest** and **compound interest** repayment options.
- Allow the user to input the **principal loan amount**, **annual interest rate**, and **loan term** in years.
- For compound interest, assume the interest compounds annually.

The formulas to use are:

- **Simple Interest Total Repayment:**

$$\text{Total} = P \times (1 + r \times t)$$

- **Compound Interest Total Repayment:**

$$\text{Total} = P \times (1 + r)^t$$

Where:

- ***P***: Principal loan amount
- ***r***: Annual interest rate (as a decimal, e.g., 5% = 0.05)
- ***t***: Loan term in years

Your program must:

1. Prompt the user to enter the **principal loan amount**, **annual interest rate**, and **loan term** in years.
2. Compute both the **simple interest** and **compound interest** repayment totals.
3. Display a detailed summary comparing the repayment totals for the two interest options.

Include Error Handling: Ensure the inputs are valid and handle invalid entries gracefully (e.g., negative or non-numeric values). **(10 Marks)**

b) Write a Python program that asks the user to input a string and a number. Then, create a new string by repeating the input string the specified number of times. Ensure ***error handling*** is in place to manage invalid inputs. **(10 Marks)**

c) Define a function ***calculate_grade(score)*** that takes a student's test score as input and returns their grade based on the following scale:

- A: 90-100
- B: 80-89
- C: 70-79
- D: 60-69
- F: Below 60

Write a program to prompt the user for their score, call the function, and display the grade. **(10 Marks)**

d) Write a Python program to create a list of numbers from 1 to 20. Iterate through the list and:

- Only print numbers divisible by 3.
- Use the break statement to stop the loop if a number exceeds 15. **(10 Marks)**

[Total: 40 Marks]

SECTION B: Attempt any THREE questions in this section

Question 2

a) Explain the concept of recursion in Python. Write a recursive function to calculate the factorial of a given number. Provide a brief explanation of how the recursion works in your program. **(10 Marks)**

b) Write a Python program that prompts the user to enter a series of numbers, one at a time. Stop taking input when the user enters a negative number. Display the average of all positive numbers entered. **(5 Marks)**

c) Define a Python function ***is_prime(num)*** that checks whether a given number is prime. Use the function to find all prime numbers between 1 and 50. **(5 Marks)**

[Total: 20 Marks]

Question 3

a) The following Python code has errors. Identify and correct the errors:

```
def find_max_number

max_number -= numbers[0]

for number in numbers;

if number < max_number

max_number += number

return max_number

print "Maximum:", max_number
```

(5 Marks)

b) Using the dictionary below, write a Python program to print the names of employees earning more than 50,000:

Salaries = {"Alice": 55000, "Bob": 48000, "Charlie": 62000, "Diana": 59000}

(5 Marks)

c) Write a Python program that calculates both the area and perimeter of a rectangle. The program should prompt the user to enter the rectangle's length and width, ***validate the input***, and then display the results. Include ***error handling*** to manage non-numeric inputs.

(10 Marks)

[Total: 20 Marks]

Question 4

a) Explain what a Python class is and why it is useful. Create a Python class named Product with the following attributes and method:

- Attributes: name, price, and quantity
- Method: ***total_value()*** that calculates and returns the total value of the product (price multiplied by quantity).

- Method: ***apply_discount(discount_percentage)*** that reduces the price by a given discount percentage

Demonstrate how to create an instance of the Product class, set its attributes, apply a discount, and call the ***total_value()*** method to display the product's total value before and after applying the discount. **(10 Marks)**

b) Write a Python program that opens a text file named records.txt. If the file does not exist, your program should create it and write the sentence "New record created." If the file does exist, your program should append the sentence "Record updated." After updating the file, your program should count the total number of lines in the file and display this count, along with the contents of the file. **(10 Marks)**

[Total: 20 Marks]

Question 5

a) Study the following Python code. What will be the output?

```
numbers = [5, 10, 15, 20, 25]

result = 1

for num in numbers:

  if num % 5 == 0:

    result *= num

  else:

    result += num

print(result)
```

(4 Marks)

b) Create a Python class Student with attributes name, age, and grades (a list of scores). Add methods to:

- Calculate and return the average grade.
- Add a new grade to the grades list.

Demonstrate the use of this class by creating an instance and calling the methods.

(8 Marks)

c) Write a Python program that simulates a number guessing game. The program should:

- Generate a random number between 1 and 100.
- Allow the user to guess the number, providing feedback after each guess ("Too high", "Too low", or "Correct").
- Keep track of the number of attempts taken to guess the number correctly.

(8 Marks)

[Total: 20 Marks]

END OF EXAMINATION